ALL UPDATED ENDPOINTS

DATABASE CONTEXT

nhi_users (Updated)

Column	Type	Purpose
id	INTEGER	Existing
email	TEXT UNIQUE	Login identifier. Used as user_id in JWT and user_settings
password	TEXT	Bcrypt hash. Never in JWT
full_name	TEXT	Display name
department	TEXT	Team/department
role	TEXT	admin, viewer, etc. Goes into JWT
tenant_id	UUID	**NEW** → nhi_global.tenants.tenant_id
status	TEXT	**NEW** active / inactive / suspended
last_login_at	TIMESTAMPTZ	**NEW**
created_at	TIMESTAMPTZ	Existing

nhi_global.tenants (Existing — untouched)

Column	Type
tenant_id	UUID PK
db_name	TEXT UNIQUE

JWT Payload

JSON

```
{
  "user_id": "abhishek.tiwari@gmail.com",
  "tenant_id": "a1b2c3d4-...",
  "db_name": "nhi_tenant_acme",
  "role": "admin",
  "exp": 1727000000
}
```

Filter Resolution (Every Data Endpoint)

Priority	Source	Logic
1	user_settings	If row exists for user_id, use saved filters
2	accounts	If no user_settings, use is_default = true rows
3	Scan	If selected_scan_id is NULL, use latest completed scan

---

ENDPOINT 1A: Get Settings

Field	Detail
Endpoint	GET /api/v1/settings
Method	GET
Features	Load user's filter preferences. Falls back to accounts defaults
Auth	JWT → user_id, tenant_id, db_name

Tables & Columns:

Table	Columns
user_settings	user_id, selected_account_ids, selected_regions, selected_scan_id
accounts	tenant_id, account_id, region, is_default
nhi_scans_aws	scan_id, scan_end, status

Response:

JSON


```
{
  "settings": {
    "accounts": {
      "selected": ["123456789012", "987654321098"],
      "is_default": true
    },
    "regions": {
      "selected": ["us-east-1", "eu-west-1"],
      "is_default": true
    },
    "scan": {
      "selected_scan_id": null,
      "effective_scan_id": "scan-2025-09-19-001",
      "effective_scan_time": "2025-09-19T08:00:00Z",
```

```
    "is_default": true
   }
  },
  "source": "accounts"
}
```

ENDPOINT 1B: Update Settings

Field	Detail
Endpoint	PUT /api/v1/settings
Method	PUT
Features	Save preferences. Upserts user_settings. JWT unchanged
Auth	JWT → user_id

Request Body:

JSON

```
{
  "selected_account_ids": ["123456789012"],
  "selected_regions": ["us-east-1"],
  "selected_scan_id": "scan-2025-09-18-002"
}
```

Tables & Columns:

Table	Columns
user_settings	user_id, selected_account_ids, selected_regions, selected_scan_id, updated_at

Response:

JSON

```
{
  "success": true,
  "settings": {
    "selected_account_ids": ["123456789012"],
    "selected_regions": ["us-east-1"],
    "selected_scan_id": "scan-2025-09-18-002"
  }
}
```

ENDPOINT 2: Filter Options

Field	Detail
Endpoint	GET /api/v1/filters
Method	GET
Features	All dropdown options for filter bars
Auth	JWT → tenant_id, db_name

Tables & Columns:

Table	Columns
identities	account_id, region, identity_type, functional_class, environment, owner, status, vpc_id
credentials	account_id, region, credential_type, status
nhi_scans_aws	scan_id, scan_end, status, total_identities, total_credentials
accounts	account_id, region, is_default

Response:

JSON


```
{
 "accounts": [
  { "account_id": "123456789012", "identity_count": 342, "credential_count": 128, "is_default": true },
  { "account_id": "987654321098", "identity_count": 87, "credential_count": 45, "is_default": false }
 ],
 "regions": [
  { "region": "us-east-1", "count": 289 },
  { "region": "eu-west-1", "count": 140 }
 ],
 "scans": [
  { "scan_id": "scan-2025-09-19-001", "scan_end": "2025-09-19T08:00:00Z", "status": "COMPLETED", "total_identities": 429,
"total_credentials": 173 }
 ],
 "identity_types": [
  { "type": "ec2_instance", "count": 120 },
  { "type": "lambda_function", "count": 95 }
 ],
 "functional_classes": [
  { "class": "service", "count": 200 },
  { "class": "application", "count": 110 }
```

```
 ],
 "credential_types": [
  { "type": "iam_role", "count": 130 },
  { "type": "access_key", "count": 28 }
 ],
 "environments": [
  { "environment": "production", "count": 180 },
  { "environment": "staging", "count": 95 }
 ],
 "statuses": [
  { "status": "active", "count": 380 },
  { "status": "stopped", "count": 35 }
 ]
}
```

ENDPOINT 3: Dashboard Overview

Field	Detail
Endpoint	GET /api/v1/dashboard/overview
Method	GET
Features	All KPIs + all charts
Auth	JWT → user_id, tenant_id, db_name

Tables & Columns:

Table	Columns
identities	id, identity_type, functional_class, account_id, region, environment, owner, status, is_publicly_accessible, first_seen_at, last_activity_at, deleted_at
credentials	id, credential_type, account_id, region, status, expires_at, last_used_at, deleted_at
identity_credentials	identity_id, credential_id
effective_permissions	credential_id, granted_count
permission_usage	credential_id, usage_count_90d
nhi_scans_aws	scan_id, scan_end, status

Response:

JSON

```
{
  "kpis": {
```

```
  "total_identities": 429,
  "total_credentials": 173,
  "orphaned_credentials": 12,
  "stale_identities": 34,
  "stale_credentials": 8,
  "publicly_accessible": 5,
  "shadow_identities": 22,
  "accounts_scanned": 3,
  "last_scan_time": "2025-09-19T08:00:00Z",
  "permission_utilization_pct": 24.3
},
"charts": {
  "identities_by_type": [
    { "type": "ec2_instance", "count": 120 },
    { "type": "lambda_function", "count": 95 }
  ],
  "identities_by_functional_class": [
    { "class": "service", "count": 200 },
    { "class": "application", "count": 110 }
  ],
  "identities_by_account": [
    { "account_id": "123456789012", "count": 342 },
    { "account_id": "987654321098", "count": 87 }
  ],
  "identities_by_region": [
    { "region": "us-east-1", "count": 289 },
    { "region": "eu-west-1", "count": 140 }
  ],
  "identities_by_environment": [
    { "environment": "production", "count": 180 },
    { "environment": "staging", "count": 95 }
  ],
  "identities_by_status": [
    { "status": "active", "count": 380 },
    { "status": "stopped", "count": 35 }
  ],
```

```
  "identity_growth": [
   { "date": "2025-07-01", "count": 310 },
   { "date": "2025-08-01", "count": 378 }
  ],
  "credential_growth": [
   { "date": "2025-07-01", "count": 120 },
   { "date": "2025-08-01", "count": 148 }
  ],
  "stale_vs_active": { "active": 395, "stale": 34 },
  "credential_types": [
   { "type": "iam_role", "count": 130 },
   { "type": "access_key", "count": 28 }
  ],
  "top_owners": [
   { "owner": "platform-team", "count": 85 },
   { "owner": "data-team", "count": 62 }
  ],
  "credentials_expiring_soon": [
   { "credential_name": "tls-cert-prod", "expires_at": "2025-09-26T00:00:00Z", "days_remaining": 7 }
  ],
  "scan_history": [
   { "scan_id": "scan-2025-09-19-001", "scan_end": "2025-09-19T08:00:00Z", "status": "COMPLETED", "total_identities": 429,
"total_credentials": 173 }
  ]
 }
}
```

ENDPOINT 4: Identities List

Field	Detail
Endpoint	GET /api/v1/identities
Method	GET
Features	Paginated, searchable, filterable, sortable identity list
Auth	JWT → user_id, tenant_id, db_name

Query Params:

Param	Type	Default	Purpose
page	integer	1	Page number
page_size	integer	25	Rows per page (max 100)
search	string	—	Search name, arn, tags
identity_type	string (comma-sep)	—	Filter by type(s)
functional_class	string (comma-sep)	—	Filter by class(es)
environment	string (comma-sep)	—	Filter by env(s)
status	string (comma-sep)	—	Filter by status(es)
owner	string	—	Filter by owner
vpc_id	string	—	Filter by VPC
is_publicly_accessible	boolean	—	Public/private
min_idle_days	integer	—	Idle days >= N
sort_by	string	last_activity_at	Sort column
sort_order	string	desc	asc or desc

Tables & Columns:

Table	Columns
identities	id, arn, name, identity_type, functional_class, account_id, region, environment, owner, status, is_publicly_accessible, vpc_id, last_activity_at, created_at_aws, cloudtrail_creator_arn, tags
identity_credentials	identity_id (COUNT)
effective_permissions	credential_id, granted_count
permission_usage	credential_id, action (COUNT used)

Response:

JSON


```
{
  "pagination": { "page": 1, "page_size": 25, "total_count": 429, "total_pages": 18 },
  "data": [
    {
      "id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
      "name": "web-prod-01",
      "arn": "arn:aws:ec2:us-east-1:123456789012:instance/i-0a1b2c3d4e5f67890",
      "identity_type": "ec2_instance",
      "functional_class": "application",
      "account_id": "123456789012",
      "region": "us-east-1",
```

```
    "environment": "production",
    "owner": "platform-team",
    "status": "active",
    "is_publicly_accessible": false,
    "vpc_id": "vpc-0abc123def456789",
    "bound_credentials_count": 1,
    "last_activity_at": "2025-09-19T14:21:00Z",
    "created_at_aws": "2025-01-10T09:00:00Z",
    "age_days": 252,
    "idle_days": 0,
    "created_by": "arn:aws:iam::123456789012:user/admin",
    "permission_utilization_pct": 24
  }
 ]
}
```

ENDPOINT 5: Identity Detail

Field	Detail
Endpoint	GET /api/v1/identities/:id
Method	GET
Features	Full identity detail. Tabs: overview, permissions, activity, relationships, credentials, raw
Auth	JWT → user_id, tenant_id, db_name

Query Params:

Param	Type	Default	Purpose
tab	string	overview	Which tab
time_range	string	90d	7d, 14d, 30d, 90d, custom
start_date	date	—	Custom range
end_date	date	—	Custom range
service	string	—	Filter by service
used_only	boolean	—	Used only
depth	integer	1	Hops: 1, 2, 3
edge_type	string	—	Filter edge type

?tab=overview

Tables: identities (all 28), identity_credentials, credentials, effective_permissions (granted_count), permission_usage

JSON


```
{
  "tab": "overview",
  "identity": {
    "id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
    "name": "web-prod-01",
    "arn": "arn:aws:ec2:us-east-1:123456789012:instance/i-0a1b2c3d4e5f67890",
    "identity_type": "ec2_instance",
    "functional_class": "application",
    "category": "Compute",
    "account_id": "123456789012",
    "region": "us-east-1",
    "vpc_id": "vpc-0abc123def456789",
    "subnet_id": "subnet-0abc123",
    "security_group_ids": ["sg-0abc123", "sg-0def456"],
    "environment": "production",
    "owner": "platform-team",
    "status": "active",
    "is_publicly_accessible": false,
    "created_at_aws": "2025-01-10T09:00:00Z",
    "age_days": 252,
    "first_seen_at": "2025-01-10T10:00:00Z",
    "last_seen_at": "2025-09-19T08:00:00Z",
    "last_activity_at": "2025-09-19T14:21:00Z",
    "idle_days": 0,
    "parent_arn": null,
    "created_by": "arn:aws:iam::123456789012:user/admin"
  },
  "discovery": {
    "discovery_source": "ec2:DescribeInstances",
    "credential_resolution": "IAM instance profile association",
    "scan_id": "scan-2025-09-19-001",
    "last_scanned": "2025-09-19T08:00:00Z",
```

```
  "data_sources": "AWS API + CloudTrail correlation"
},
"type_attributes": {
  "instance_id": "i-0a1b2c3d4e5f67890",
  "ami_id": "ami-0abcdef1234567890",
  "instance_type": "t3.large",
  "platform": "linux",
  "key_pair": "prod-key",
  "iam_instance_profile": "arn:aws:iam::123456789012:instance-profile/ec2-app-profile",
  "ebs_volume_count": 2,
  "eni_ids": ["eni-0abc123"],
  "monitoring_state": "enabled",
  "architecture": "x86_64"
},
"bound_credentials": [
 {
   "credential_id": "b2c3d4e5-f6a7-8901-bcde-f12345678901",
   "name": "ec2-app-runtime-role",
   "credential_type": "iam_role",
   "binding_type": "instance_profile",
   "is_primary": true,
   "trust_principal": "ec2.amazonaws.com",
   "attached_policies_count": 2,
   "inline_policies_count": 2,
   "granted_actions_count": 186,
   "used_actions_count": 44,
   "utilization_pct": 24
 }
],
"tags": { "Name": "web-prod-01", "Environment": "production", "Owner": "platform-team", "Team": "backend" },
"missing_tags": ["CostCenter", "Project"],
"risk_signals": [
 { "signal": "Publicly accessible", "status": false, "severity": "critical" },
 { "signal": "Stale (no activity > 90d)", "status": false, "severity": "high" },
 { "signal": "No owner attribution", "status": false, "severity": "medium" },
 { "signal": "Credential never rotated", "status": true, "severity": "high" },
```

```
  { "signal": "Credential expired/expiring", "status": false, "severity": "critical" },
  { "signal": "Cross-account trust", "status": false, "severity": "medium" },
  { "signal": "Wildcard resource scope", "status": true, "severity": "high" },
  { "signal": "No VPC", "status": false, "severity": "medium" }
 ]
}
```

?tab=permissions

Tables: identity_credentials, effective_permissions (permissions JSONB — unnested here), permission_usage, credentials

Internal: Only endpoint that unnests jsonb_array_elements(permissions). Joins with permission_usage for used/unused.

JSON


```
{
 "tab": "permissions",
 "summary": {
  "granted_count": 186,
  "used_count": 44,
  "unused_count": 142,
  "utilization_pct": 24
 },
 "permissions": [
  {
   "service": "s3",
   "action": "s3:GetObject",
   "resource_arn": "*",
   "effect": "Allow",
   "policy_source": "AmazonS3FullAccess",
   "policy_type": "managed",
   "used": true,
   "last_used_at": "2025-09-19T14:21:00Z",
   "source_ips": ["10.0.1.42"]
  },
  {
   "service": "s3",
   "action": "s3:DeleteObject",
   "resource_arn": "*",
```

```
    "effect": "Allow",
    "policy_source": "AmazonS3FullAccess",
    "policy_type": "managed",
    "used": false,
    "last_used_at": null,
    "source_ips": []
   }
 ],
 "policy_breakdown": [
  { "policy_name": "AmazonS3FullAccess", "policy_type": "managed", "total_permissions": 45, "used_permissions": 3 },
  { "policy_name": "CustomAppPolicy", "policy_type": "inline", "total_permissions": 12, "used_permissions": 8 }
 ]
}
```

?tab=activity

Tables: identity_credentials, identity_usage

JSON


```
{
 "tab": "activity",
 "time_range": "14d",
 "volume_chart": [
  { "date": "2025-09-05", "count": 85 },
  { "date": "2025-09-06", "count": 102 }
 ],
 "recent_events": [
  {
    "event_time": "2025-09-19T14:21:00Z",
    "event_name": "GetObject",
    "service": "s3",
    "resource_arn": "arn:aws:s3:::web-assets-prod/logo.png",
    "source_ip": "10.0.1.42",
    "region": "us-east-1",
    "user_agent": "aws-sdk-java/2.17",
    "error_code": null,
    "request_id": "ABCD1234EFGH5678"
```

```
  }
 ],
 "analytics": {
  "source_ips": [{ "ip": "10.0.1.42", "pct": 100 }],
  "active_regions": [{ "region": "us-east-1", "count": 320 }],
  "top_services": [
   { "service": "s3", "pct": 75, "count": 240 },
   { "service": "ssm", "pct": 25, "count": 80 }
  ],
  "top_actions": [
   { "action": "GetObject", "count": 180 },
   { "action": "PutObject", "count": 60 }
  ],
  "error_rate_pct": 1.2,
  "access_denied_rate_pct": 0.8,
  "peak_hours": [{ "hour": 14, "count": 45 }],
  "usage_pattern": "Continuous"
 },
 "error_events": [
  {
   "event_time": "2025-09-19T12:00:00Z",
   "event_name": "PutObject",
   "service": "s3",
   "resource_arn": "arn:aws:s3:::restricted-bucket/data",
   "source_ip": "10.0.1.42",
   "error_code": "AccessDenied",
   "error_message": "User is not authorized to perform s3:PutObject"
  }
 ],
 "events_pagination": { "page": 1, "page_size": 50, "total_count": 320, "total_pages": 7 }
}
```

?tab=relationships

Tables: trust_edges, identities, credentials, resources

JSON

```
{
  "tab": "relationships",
  "graph": {
    "nodes": [
      { "arn": "arn:aws:ec2:us-east-1:123456789012:instance/i-0a1b2c3d", "name": "web-prod-01", "type": "identity", "subtype": "ec2_instance", "is_current": true },
      { "arn": "arn:aws:iam::123456789012:role/ec2-app-runtime-role", "name": "ec2-app-runtime-role", "type": "credential", "subtype": "iam_role", "is_current": false },
      { "arn": "arn:aws:s3:::web-assets-prod", "name": "web-assets-prod", "type": "resource", "subtype": "s3_bucket", "is_current": false }
    ],
    "edges": [
      { "source": "arn:aws:ec2:...:i-0a1b2c3d", "target": "arn:aws:iam::...:role/ec2-app-runtime-role", "edge_type": "BOUND_TO", "last_observed_at": "2025-09-19T08:00:00Z" },
      { "source": "arn:aws:ec2:...:i-0a1b2c3d", "target": "arn:aws:s3:::web-assets-prod", "edge_type": "ACCESSES", "last_observed_at": "2025-09-19T14:21:00Z", "metadata": { "call_count_90d": 180 } }
    ]
  },
  "relationship_table": [
    { "relationship": "Bound to", "direction": "outbound", "target_arn": "arn:aws:iam::...:role/ec2-app-runtime-role", "target_name": "ec2-app-runtime-role", "target_type": "iam_role", "last_active": "2025-09-19T08:00:00Z" },
    { "relationship": "Accesses", "direction": "outbound", "target_arn": "arn:aws:s3:::web-assets-prod", "target_name": "web-assets-prod", "target_type": "s3_bucket", "last_active": "2025-09-19T14:21:00Z" }
  ]
}
```

?tab=credentials

Tables: identity_credentials, credentials (all 31), effective_permissions (granted_count, permissions), permission_usage

JSON


```
{
  "tab": "credentials",
  "credentials": [
    {
      "credential_id": "b2c3d4e5-f6a7-8901-bcde-f12345678901",
      "name": "ec2-app-runtime-role",
      "arn": "arn:aws:iam::123456789012:role/ec2-app-runtime-role",
      "credential_type": "iam_role",
```

"binding_type": "instance_profile",

"binding_source": "ec2:DescribeInstances",

"is_primary": true,

"trust_policy": { "Version": "2012-10-17", "Statement": [{ "Effect": "Allow", "Principal": { "Service": "ec2.amazonaws.com" }, "Action": "sts:AssumeRole" }] },

"trust_principals": [{ "principal": "ec2.amazonaws.com", "type": "Service", "condition": null, "is_external": false }],

"attached_policies": [{ "name": "AmazonS3FullAccess", "type": "AWS Managed", "arn": "arn:aws:iam::aws:policy/AmazonS3FullAccess", "permissions_count": 45 }],

"inline_policies": [{ "name": "app-specific", "statements": 3, "actions": 12 }],

"lifecycle": {

"created_at_aws": "2025-01-10T09:00:00Z",

"last_used_at": "2025-09-19T14:21:00Z",

"last_rotated_at": "2025-06-15T00:00:00Z",

"age_days": 252,

"days_since_rotation": 97,

"rotation_enabled": false,

"rotation_interval_days": null,

"rotation_overdue": false,

"expires_at": null,

"max_session_duration": 3600,

"permission_boundary_arn": null

},

"assumable_roles": [{ "target_role_arn": "arn:aws:iam::987654321098:role/cross-account-reader", "target_account": "987654321098", "conditions": "ExternalId required", "last_assumed": "2025-09-05T10:00:00Z" }]

}

]

}

?tab=raw

Tables: identities (all 28), identity_credentials, credentials (raw_api_response, trust_policy, attached_policies, inline_policies, type_attributes), effective_permissions (permissions)

JSON

{

"tab": "raw",

"identity_record": { "id": "a1b2c3d4-...", "arn": "...", "name": "web-prod-01", "...": "all 28 columns" },

"raw_api_response": { "Reservations": [{ "Instances": [{ "InstanceId": "i-0a1b2c3d", "InstanceType": "t3.large" }] }] },

"resolved_credential": { "roleArn": "arn:aws:iam::...:role/ec2-app-runtime-role", "assumedVia": "instance profile", "attachedPolicies": 2, "inlinePolicies": 2 },

"type_attributes": { "instance_id": "i-0a1b2c3d", "ami_id": "ami-0abcdef", "instance_type": "t3.large" }

}

ENDPOINT 6: Credentials List

Field	Detail
Endpoint	GET /api/v1/credentials
Method	GET
Features	Paginated, searchable, filterable, sortable credentials list
Auth	JWT → user_id, tenant_id, db_name

Query Params:

Param	Type	Default	Purpose
page	integer	1	Page
page_size	integer	25	Per page
search	string	—	name, arn
credential_type	string (comma-sep)	—	Type(s)
status	string (comma-sep)	—	Status(es)
owner_principal_arn	string	—	Owner
rotation_enabled	boolean	—	Rotation
rotation_overdue	boolean	—	Overdue
is_orphaned	boolean	—	Orphaned
has_external_trust	boolean	—	External trust
min_idle_days	integer	—	Idle >= N
max_days_until_expiry	integer	—	Expiring within N
sort_by	string	last_used_at	Sort
sort_order	string	desc	Order

Tables: credentials, identity_credentials

Response:

JSON

{

"pagination": { "page": 1, "page_size": 25, "total_count": 173, "total_pages": 7 },

"data": [

```
  {
    "id": "b2c3d4e5-f6a7-8901-bcde-f12345678901",
    "name": "ec2-app-runtime-role",
    "arn": "arn:aws:iam::123456789012:role/ec2-app-runtime-role",
    "credential_type": "iam_role",
    "account_id": "123456789012",
    "region": "global",
    "status": "active",
    "owner_principal_arn": "arn:aws:iam::123456789012:root",
    "bound_identities_count": 3,
    "is_orphaned": false,
    "last_used_at": "2025-09-19T14:21:00Z",
    "last_rotated_at": "2025-06-15T00:00:00Z",
    "rotation_enabled": false,
    "rotation_overdue": false,
    "age_days": 252,
    "idle_days": 0,
    "expires_at": null,
    "days_until_expiry": null,
    "has_external_trust": false,
    "permission_boundary_arn": null
  }
 ]
}
```

ENDPOINT 7: Credential Detail

Field	Detail
Endpoint	GET /api/v1/credentials/:id
Method	GET
Features	Full credential detail. Tabs: overview, trust, activity, rotation, relationships, raw
Auth	JWT → user_id, tenant_id, db_name

Query Params:

Param	Type	Default	Purpose
tab	string	overview	Which tab
time_range	string	90d	Activity
depth	integer	1	Relationships
edge_type	string	—	Relationships

?tab=overview

Tables: credentials (all 31), identity_credentials, identities

JSON


```
{
  "tab": "overview",
  "credential": {
    "id": "b2c3d4e5-f6a7-8901-bcde-f12345678901",
    "name": "ec2-app-runtime-role",
    "arn": "arn:aws:iam::123456789012:role/ec2-app-runtime-role",
    "credential_type": "iam_role",
    "category": "IAM",
    "account_id": "123456789012",
    "region": "global",
    "status": "active",
    "owner_principal_arn": "arn:aws:iam::123456789012:root",
    "created_at_aws": "2025-01-10T09:00:00Z",
    "age_days": 252,
    "last_used_at": "2025-09-19T14:21:00Z",
    "idle_days": 0,
    "last_rotated_at": "2025-06-15T00:00:00Z",
    "days_since_rotation": 97,
    "rotation_enabled": false,
    "rotation_interval_days": null,
    "rotation_status": "Disabled",
    "expires_at": null,
    "days_until_expiry": null,
    "max_session_duration": 3600,
    "permission_boundary_arn": null
  },
```

```
 "bound_identities": [
  { "identity_id": "a1b2c3d4-...", "name": "web-prod-01", "identity_type": "ec2_instance", "account_id": "123456789012", "status": "active", "last_activity_at": "2025-09-19T14:21:00Z", "binding_type": "instance_profile" }
 ],
 "risk_signals": [
  { "signal": "Orphaned (no identity bound)", "status": false, "severity": "high" },
  { "signal": "Overshared (3+ identities)", "status": false, "severity": "medium" },
  { "signal": "Never rotated", "status": true, "severity": "high" },
  { "signal": "Rotation overdue", "status": false, "severity": "high" },
  { "signal": "Expiring soon (< 30 days)", "status": false, "severity": "high" },
  { "signal": "Already expired", "status": false, "severity": "critical" },
  { "signal": "Stale (unused > 90 days)", "status": false, "severity": "high" },
  { "signal": "External trust (cross-account)", "status": false, "severity": "medium" },
  { "signal": "Wildcard trust (Principal: *)", "status": false, "severity": "critical" },
  { "signal": "No condition on trust policy", "status": true, "severity": "high" },
  { "signal": "No permission boundary", "status": true, "severity": "medium" },
  { "signal": "Wildcard resource in policies", "status": true, "severity": "high" }
 ],
 "type_attributes": { "trust_policy_summary": "1 service principal", "instance_profile_arn": "arn:aws:iam::123456789012:instance-profile/ec2-app-profile", "service_linked": false, "path": "/" }
}
```

?tab=trust

Tables: credentials, effective_permissions (permissions JSONB — unnested for summary), permission_usage

JSON


```
{
 "tab": "trust",
 "trust_policy": {
  "raw": { "Version": "2012-10-17", "Statement": [{ "Effect": "Allow", "Principal": { "Service": "ec2.amazonaws.com" }, "Action": "sts:AssumeRole" }] },
  "principals": [{ "principal": "ec2.amazonaws.com", "type": "Service", "condition": null, "is_external": false }],
  "warnings": []
 },
 "attached_policies": [
  { "name": "AmazonS3FullAccess", "type": "AWS Managed", "arn": "arn:aws:iam::aws:policy/AmazonS3FullAccess", "permissions_count": 45, "managed_by": "AWS" }
```

```
 ],
 "inline_policies": [
  { "name": "app-specific", "statements": 3, "actions": 12, "resources": "Scoped" }
 ],
 "effective_permissions_summary": [
  { "service": "s3", "allowed_actions": 45, "denied_actions": 0, "resource_scope": "*" },
  { "service": "ssm", "allowed_actions": 22, "denied_actions": 0, "resource_scope": "arn:aws:ssm:*:*:parameter/app/*" }
 ],
 "assumable_roles": [
  { "target_role_arn": "arn:aws:iam::987654321098:role/cross-account-reader", "target_account": "987654321098", "conditions": "ExternalId required", "last_assumed": "2025-09-05T10:00:00Z" }
 ],
 "permission_boundary": { "boundary_arn": null, "effective_restriction": null, "actions_blocked": [] }
}
```

?tab=activity

Tables: identity_usage, identities

JSON


```
{
 "tab": "activity",
 "time_range": "90d",
 "volume_chart": [{ "date": "2025-06-21", "count": 120 }],
 "recent_events": [
  { "event_time": "2025-09-19T14:21:00Z", "event_name": "GetObject", "service": "s3", "resource_arn": "arn:aws:s3:::web-assets-prod/logo.png", "source_ip": "10.0.1.42", "region": "us-east-1", "user_agent": "aws-sdk-java/2.17", "error_code": null }
 ],
 "identities_using_this_credential": [
  { "identity_id": "a1b2c3d4-...", "name": "web-prod-01", "identity_type": "ec2_instance", "source_ip": "10.0.1.42", "last_action": "s3:GetObject", "last_action_time": "2025-09-19T14:21:00Z" }
 ],
 "analytics": {
  "source_ips": [{ "ip": "10.0.1.42", "pct": 60 }, { "ip": "10.0.1.43", "pct": 40 }],
  "active_regions": [{ "region": "us-east-1", "count": 500 }],
  "top_services": [{ "service": "s3", "pct": 75, "count": 375 }],
  "usage_pattern": "Continuous",
  "error_rate_pct": 1.0
```

```
 },
 "events_pagination": { "page": 1, "page_size": 50, "total_count": 500, "total_pages": 10 }
}
```

?tab=rotation

Tables: credentials

JSON


```
{
 "tab": "rotation",
 "timeline": [
  { "date": "2025-01-10T09:00:00Z", "event": "Created", "details": "By admin@company.com" },
  { "date": "2025-03-10T00:00:00Z", "event": "Rotated", "details": "Manual" },
  { "date": "2025-06-15T00:00:00Z", "event": "Rotated", "details": "Manual" }
 ],
 "configuration": {
  "auto_rotation": false,
  "rotation_interval_days": null,
  "last_rotation": "2025-06-15T00:00:00Z",
  "next_expected": null,
  "status": "Disabled",
  "rotation_lambda_arn": null
 },
 "lifecycle": {
  "created_at_aws": "2025-01-10T09:00:00Z",
  "age_days": 252,
  "days_since_rotation": 97,
  "expires_at": null,
  "days_until_expiry": null
 }
}
```

?tab=relationships

Tables: trust_edges, identities, credentials, resources

JSON

```
{
  "tab": "relationships",
  "graph": {
    "nodes": [
      { "arn": "arn:aws:iam::...:role/ec2-app-runtime-role", "name": "ec2-app-runtime-role", "type": "credential", "subtype": "iam_role", "is_current": true },
      { "arn": "arn:aws:ec2:...:i-0a1b2c3d", "name": "web-prod-01", "type": "identity", "subtype": "ec2_instance", "is_current": false }
    ],
    "edges": [
      { "source": "arn:aws:ec2:...:i-0a1b2c3d", "target": "arn:aws:iam::...:role/ec2-app-runtime-role", "edge_type": "BOUND_TO", "last_observed_at": "2025-09-19T08:00:00Z" }
    ]
  },
  "relationship_table": [
    { "relationship": "Used by", "direction": "inbound", "target_arn": "arn:aws:ec2:...:i-0a1b2c3d", "target_name": "web-prod-01", "target_type": "ec2_instance", "last_active": "2025-09-19T14:21:00Z" }
  ]
}
```

?tab=raw

Tables: credentials (all 31), effective_permissions (permissions)

JSON


```
{
  "tab": "raw",
  "credential_record": { "id": "b2c3d4e5-...", "...": "all 31 columns" },
  "raw_api_response": { "Role": { "RoleName": "ec2-app-runtime-role", "RoleId": "AROA123456789EXAMPLE" } },
  "trust_policy": { "Version": "2012-10-17", "Statement": ["..."] },
  "attached_policies": [{ "PolicyName": "AmazonS3FullAccess", "Document": { "..." } }],
  "inline_policies": [{ "PolicyName": "app-specific", "Document": { "..." } }],
  "type_attributes": { "instance_profile_arn": "...", "service_linked": false, "path": "/" },
  "effective_permissions": [
    { "service": "s3", "action": "s3:GetObject", "resource_arn": "*", "effect": "Allow", "policy_source": "AmazonS3FullAccess", "policy_type": "managed" }
  ]
}
```

ENDPOINT SUMMARY

#	Endpoint	Method	Features	Primary Tables
1a	/api/v1/settings	GET	Load preferences	user_settings, accounts, nhi_scans_aws
1b	/api/v1/settings	PUT	Save preferences	user_settings
2	/api/v1/filters	GET	Dropdown options	identities, credentials, nhi_scans_aws, accounts
3	/api/v1/dashboard/overview	GET	KPIs + Charts	identities, credentials, identity_credentials, effective_permissions, permission_usage, nhi_scans_aws
4	/api/v1/identities	GET	Identity list	identities, identity_credentials, effective_permissions, permission_usage
5	/api/v1/identities/:id?tab=	GET	Identity detail (6 tabs)	All tables
6	/api/v1/credentials	GET	Credential list	credentials, identity_credentials
7	/api/v1/credentials/:id?tab=	GET	Credential detail (6 tabs)	All tables

---

BACKEND TEAM: Auth + Settings Flow

Login

text

1. User sends email + password
2. SELECT id, email, full_name, role, tenant_id, status FROM nhi_users WHERE email = :email
3. Reject if status != 'active'
4. Verify bcrypt password
5. SELECT db_name FROM nhi_global.tenants WHERE tenant_id = :tenant_id
6. UPDATE nhi_users SET last_login_at = NOW() WHERE id = :id
7. Issue JWT: { user_id: email, tenant_id, db_name, role, exp }

Every Data API

text

1. Extract user_id, tenant_id, db_name from JWT
2. Connect to db_name database
3. SELECT * FROM user_settings WHERE user_id = :user_id
4. Found → use saved filters
5. Not found → SELECT DISTINCT account_id, region FROM accounts WHERE tenant_id = :tenant_id AND is_default = true
6. Scan: if NULL → latest completed from nhi_scans_aws
7. Apply WHERE account_id IN (...) AND region IN (...) AND scan_id = :scan_id

Settings Change

text

1. PUT /api/v1/settings → upsert user_settings
2. JWT unchanged
3. Next API call reads updated user_settings from DB